

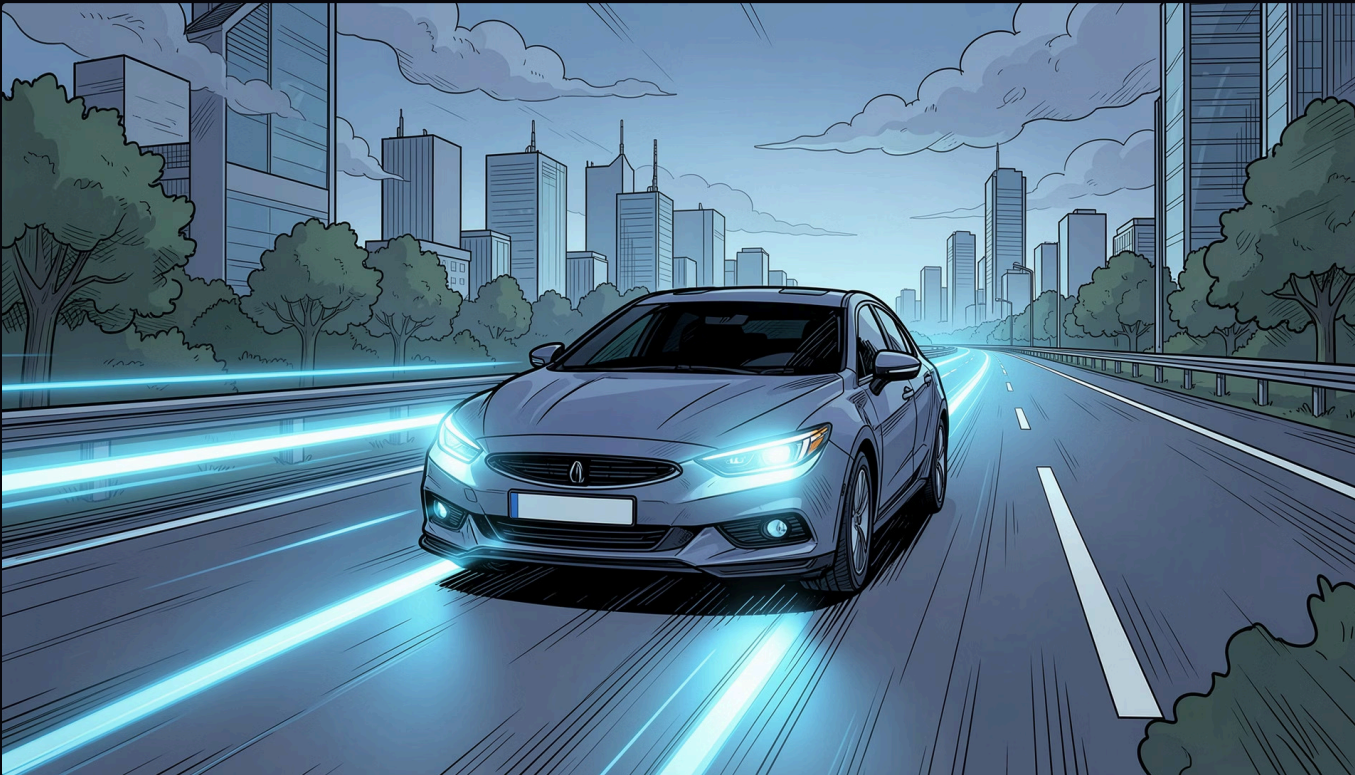


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Chapter 3: Motion in a Straight Line

Class 11 Physics — Complete Chapter Notes

Introduction to Motion



Motion is the change in position of an object with respect to time and its surroundings.

Rectilinear Motion

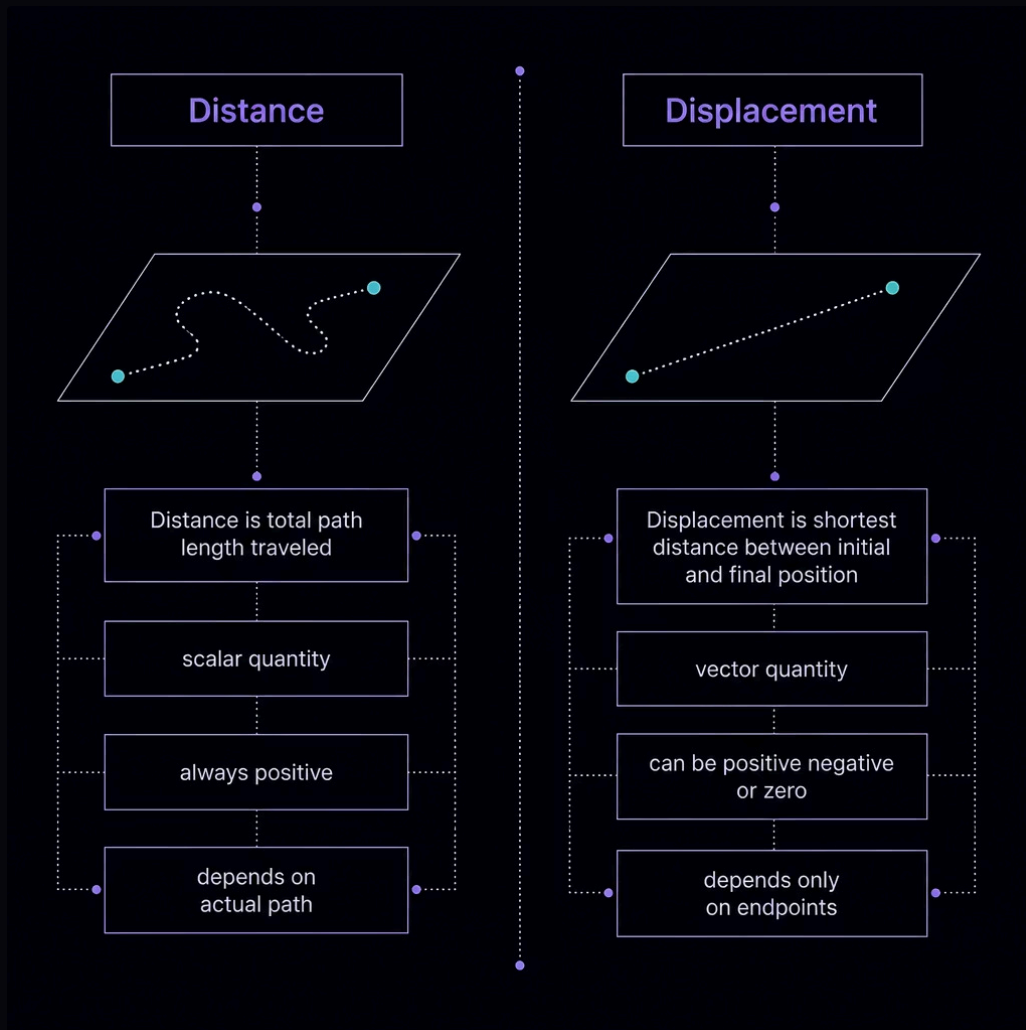
Motion along a straight line (1D motion)

Frame of Reference

A coordinate system used to measure position and motion

- ① Position is always measured relative to a chosen **origin (0)**.

Distance vs Displacement



Key Differences

- **Distance** — Total path length covered. Scalar. Always positive.
- **Displacement** — Shortest path between start and end. Vector. Can be zero or negative.

□ Displacement $\leq$ Distance always. They are equal only when motion is in one direction without turning back.

Speed and Velocity

⚡ Average Speed

Total distance $\div$ Total time

Scalar quantity

🎯 Average Velocity

Total displacement $\div$ Total time

Vector quantity

🔬 Instantaneous Velocity

Velocity at a specific instant

$$v = \lim_{\Delta t \rightarrow 0} (\Delta x / \Delta t)$$



Formula Summary

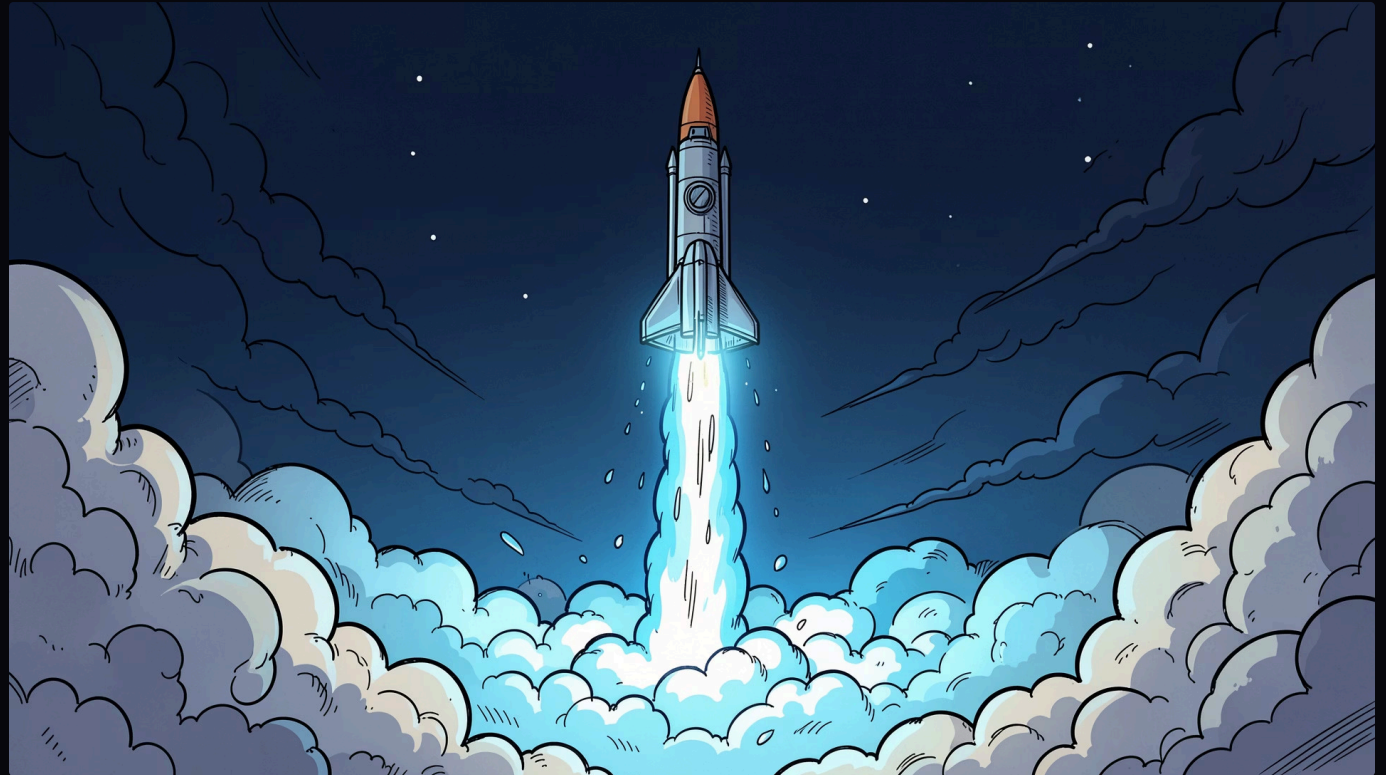
- Average Speed = **Total Distance** / **Total Time**
- Average Velocity = **Displacement** / **Time**
- Instantaneous Velocity = **dx/dt**

Acceleration

What is Acceleration?

Rate of change of velocity with time.

- **Formula:** $a = (v - u) / t$
- **SI Unit:** m/s^2
- **Nature:** Vector quantity



Positive Acceleration

Velocity increases with time

Negative Acceleration (Retardation)

Velocity decreases with time

Equations of Motion (Uniform Acceleration)

1

First Equation

$$v = u + at$$

Relates velocity, time and acceleration

2

Second Equation

$$s = ut + \frac{1}{2}at^2$$

Relates displacement, time and acceleration

3

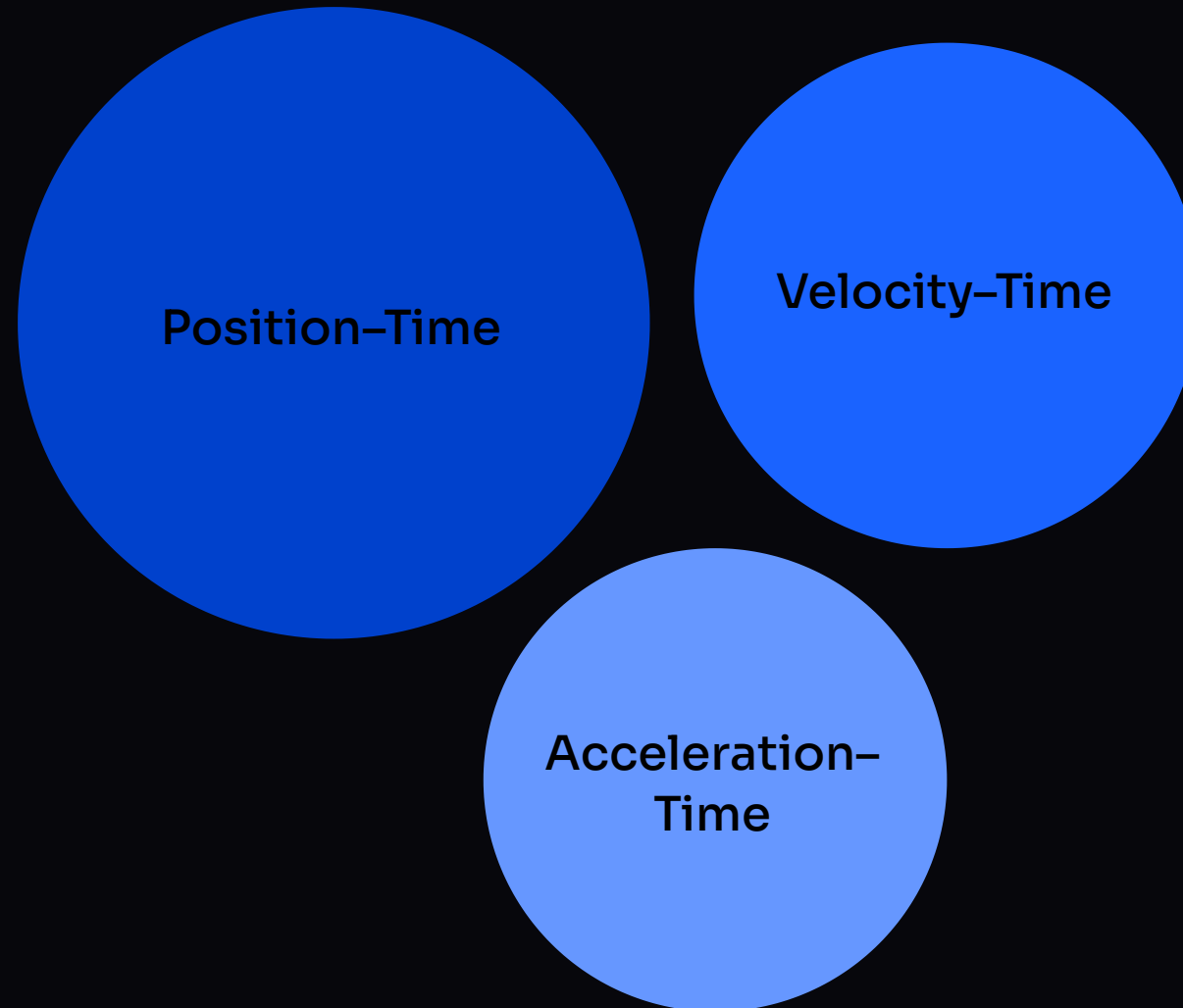
Third Equation

$$v^2 = u^2 + 2as$$

Relates velocity and displacement (time-independent)

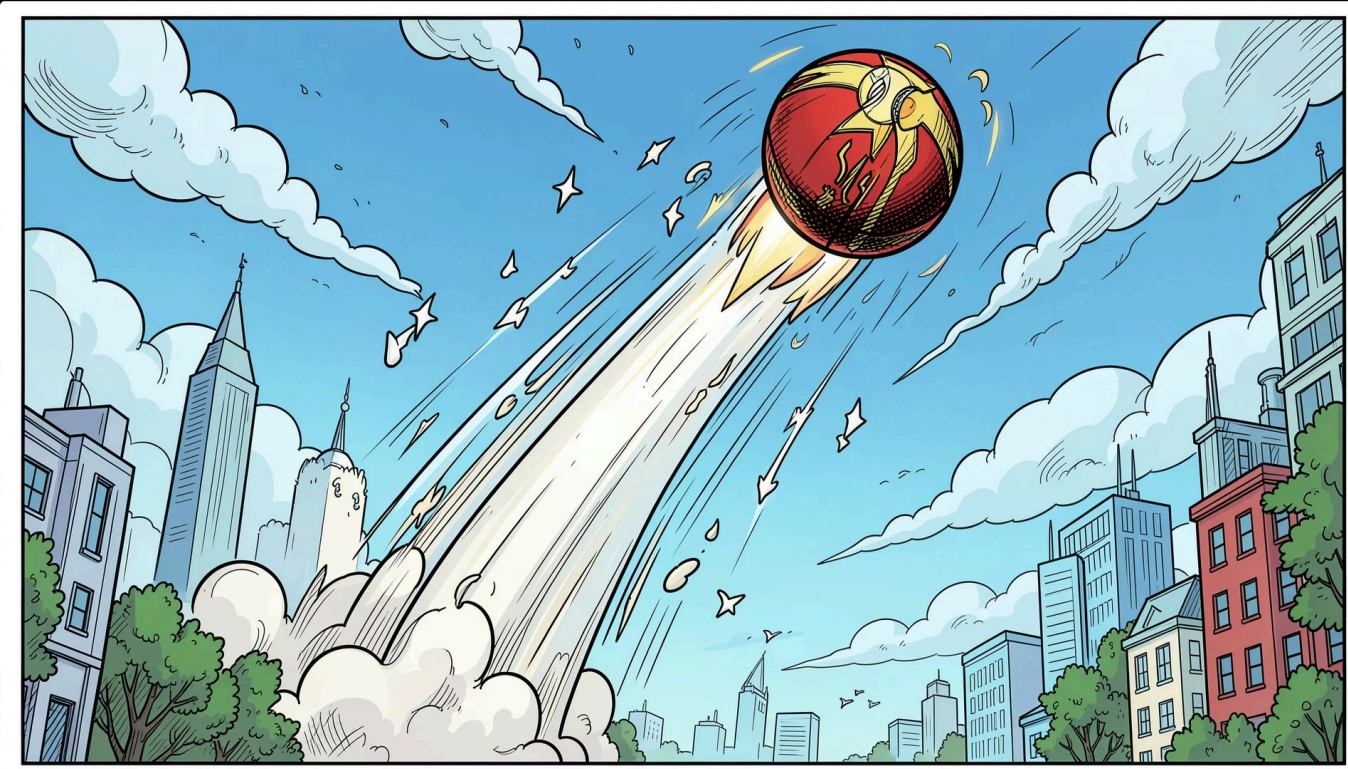
❗ Here, **u** = initial velocity, **v** = final velocity, **a** = acceleration, **s** = displacement, **t** = time

Graphical Representation of Motion



Graphs are powerful tools to visualize motion. The **slope** of a position-time graph gives velocity, while the **area under** a velocity-time graph gives displacement.

Motion Under Gravity



Key Points

- Acceleration due to gravity: $g = 9.8 \text{ m/s}^2$ (downward)
- For upward motion: $a = -g$
- At maximum height: $v = 0$
- Time of ascent = Time of descent

Modified Equations

- $v = u - gt$
- $h = ut - \frac{1}{2}gt^2$
- $v^2 = u^2 - 2gh$

Relative Velocity

Concept

Velocity of object A **with respect to** object B is called relative velocity.

- **Same direction:** $V_{AB} = V_A - V_B$
- **Opposite direction:** $V_{AB} = V_A + V_B$

❏ If two objects move with the same velocity in the same direction, their relative velocity is **zero**.

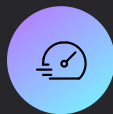


Quick Revision — Key Formulas



Displacement

$$\Delta x = x_2 - x_1$$



Average Velocity

$$\bar{v} = \Delta x / \Delta t$$



Acceleration

$$a = (v - u) / t$$



Free Fall

$$g = 9.8 \text{ m/s}^2$$



Relative Velocity

$$V_{AB} = V_A - V_B$$

✓ Master these formulas and concepts — you're ready for any exam question on Motion in a Straight Line! 🎯